Shashi Bhusan Mishra

University of Maryland, College Park, MD 20740
Phone: (646) 961-9913 ⋄ Email: shashi13@umd.edu
Google Scholar ⋄ LinkedIn ⋄ GitLab ⋄ Webpage ⋄ ORCID

RESEARCH SUMMARY

I am a theoretical and computational condensed matter physicist developing predictive first-principles approaches to understand and design quantum materials. My research connects crystal and magnetic symmetry with many-body interactions to uncover the microscopic mechanisms governing superconductivity, unconventional magnetism, topological transport, and light-driven phenomena. I combine density-functional and many-body theories with Wannier methods, spin models, and Berry-phase approaches, and contribute to the open-source EPW and EPWpy codes.

ACADEMIC APPOINTMENTS

Assistant Research Scientist March 2026 – present
Department of Electrical and Computer Engineering, University of Maryland, College Park, USA
Group Head: Prof. Thomas E. Murphy

Postdoctoral Fellow April 2023 – March 2026
Department of Physics, SUNY Binghamton, NY, USA
Advisor: Prof. Roxana Margine

Postdoctoral Fellow April 2021 – March 2023
Department of Mechanical Engineering, University of California, Riverside, USA
Advisor: Prof. Sinisa Coh

EDUCATION

Ph.D. in Physics March 2021
Indian Institute of Technology Madras, India
Thesis: Ab initio Modelling and Functionalization of Graphene & TiO_2 Surfaces and Interfaces
Advisors: Prof. B. R. K. Nanda & Prof. Somnath C. Roy

M.Sc. in Physics June 2013
Utkal University, Odisha, India

B.Sc. in Physics June 2011
Fakir Mohan University, Odisha, India

RESEARCH INTERESTS

- Magnetism and spintronics
- Superconductivity and electron–phonon physics
- Topological materials
- Transport and Berry-phase phenomena
- Light–matter interactions
- First-principles materials discovery

RESEARCH DIRECTIONS

- **2D-magnetism and multiferroicity:** Design van der Waals and oxide heterostructures where ferroelectric polarization and interfacial symmetry breaking enable electrically controlled magnetism.
- **Superconductivity:** Go beyond Migdal–Eliashberg theory with anharmonicity, vertex corrections, full-bandwidth effects, and *ab initio* Coulomb interactions, for hydrides, 2D, and noncentrosymmetric superconductors.
- **Topology and quantum transport:** Combine Berry-phase physics with first-principles transport to predict anomalous and spin Hall effects and magnetotransport in topological materials.
- **Light-driven quantum materials:** First-principles theory of optically induced magnetism and the inverse Faraday effect for light-controlled magnetic phases.
- **Data-driven materials discovery:** Combine materials databases and machine learning with first-principles calculations to find promising magnets and superconductors.

PUBLICATIONS

30 peer-reviewed articles | 17 first/co-first-author papers | 7 manuscripts under review | 508 citations | h-index 15. * Equal contribution.

1. M. Keivanloo, **S. B. Mishra**, A. Shariatmanesh, M. Sandoghchi, K. Hongo, R. Maezono, E. R. Margine, S. Esmaeili, and M. Khazaei, "Superconductivity and band topology in functionalized 2D hexagonal MBenes," *Journal of Materials Chemistry C* (2026).

2. M. Keivanloo, F. Dinmohammad, **S. B. Mishra**, M. Sandoghchi, M. J. Arshia, M. Kawamura, E. R. Margine, M. H. Mahyuddin, H. Raebiger, R. P. P. Sukanli, K. Hongo, R. Maezono, and M. Khazaei, "High-T_c superconductivity in functionalized out-of-plane ordered double transition metal MXenes," *npj Computational Materials* (2026).

3. **S. B. Mishra***, Z. Liu*, S. Tiwari, F. Giustino, and E. R. Margine, "Comparative study of phonon-limited carrier transport in the Weyl semimetal TaAs family," *Physical Review B* **113**, 045202 (2026).

4. **S. B. Mishra** and R. Margine, "Nonadiabatic and anharmonic effects in high-pressure H_3S and D_3S superconductors," *Annalen der Physik* **538**, e00553 (2026).

5. S. Dey*, **S. B. Mishra***, A. Hazra, D. Kabiraj, and S. C. Roy, "Anomalous gas sensing characteristics of TiO_2 nanorod arrays irradiated with high-energy ion beam," *Materials Science and Engineering: B* **326**, 119210 (2026).

6. **S. B. Mishra**, H. Mori, and E. R. Margine, "Electron-phonon vertex correction effect in superconducting H_3S," *npj Computational Materials* **11**, 342 (2025).

7. **S. B. Mishra**, "Inverse Faraday effect in $3d$, $4d$, and $5d$ transition metals," *Physical Review B* **111**, 174413 (2025).

8. Z. Liu*, **S. B. Mishra***, J.-M. Lihm, S. Poncé, and E. R. Margine, "Phonon-limited carrier transport in the Weyl semimetal TaAs," *Physical Review B* **112**, 104311 (2025).

9. Y. Rambabu, **S. B. Mishra**, O. Henrotte, and E. Cortés, "Defects dynamics in photo-excited CeO_2 and their influence on CO_2 photoreduction," *Advanced Functional Materials*, e13933 (2025).

10. X. Zhang, **S. B. Mishra**, E. R. Margine, and E. Kioupakis, "Cubic BeB_2: a metastable p-type conductive material from first principles," *Physical Review B* **112**, 155206 (2025).

11. A. Das, S. Gumber, N. C. Maji, **S. B. Mishra**, Preethi M., P. Ning, R. G. Nair, Abhijith T, A. S. Vasenko, Madhumitha R., and O. V. Prezhdo, "Electrostatic self-assembly driven heterojunction of CeO_2/g-C_3N_4 nanosheets for efficient photocatalytic hydrogen evolution," *ACS Applied Materials & Interfaces* (2025).

12. **S. B. Mishra**, E. T. Marcial, S. Debata, A. N. Kolmogorov, and E. R. Margine, "Stability–superconductivity map for compressed Na-intercalated graphite," *Physical Review B* **110**, 174508 (2024).

13. P. Sahoo, **S. B. Mishra**, S. Debata, A. Sharma, B. Sahu, S. Padhan, R. Thangavel, and B.-K. Lee, "Hydrothermally grown halogen-doped ZnO nanorods for photoelectrochemical water oxidation: experimental and DFT insights," *Journal of Physical Chemistry C* **128**, 18711 (2024).

14. Abhijitha V. G., **S. B. Mishra**, and B. R. K. Nanda, "Si_2BN nanosheets as anchoring cathode material for realizing high capacity Al-ion battery," *Journal of Energy Storage* **77**, 109913 (2024).

15. V. Ortiz, **S. B. Mishra**, L. Vuong, S. Coh, and R. B. Wilson, "Transient ellipsometry measurements of the specular inverse Faraday effect in transition metals," *Physical Review Materials* **7**, 125202 (2023).

16. **S. B. Mishra** and S. Coh, "Spin contribution to the inverse Faraday effect of nonmagnetic metals," *Physical Review B* **107**, 214432 (2023).

17. B. Biswal, **S. B. Mishra**, R. Yadav, S. Poudyal, R. Rajarapu, P. K. Barman, K. R. Pandurang, M. Mandal, R. Singh, B. R. K. Nanda, and A. Misra, "Work function of van der Waals topological semimetals," *Applied Physics Letters* **120**, 093101 (2022).

18. Abhijitha V. G., **S. B. Mishra**, S. Ramaprabhu, and B. R. K. Nanda, "Design of aluminium ion battery with graphyne host," *Nanoscale Advances* **4**, 3870 (2022).

19. S. Dey, A. Chakravorty, **S. B. Mishra**, N. Khatun, A. Hazra, B. R. K. Nanda, S. Chandran, D. Kabiraj, and S. C. Roy, "High energy ion induced modification in TiO_2 nanorods," *Nanoscale Advances* **4**, 241 (2022).

20. **S. B. Mishra**, Abhijitha V. G., S. Ramaprabhu, and B. R. K. Nanda, "Graphdiyne—a promising cathode material for aluminium dual ion battery," *ACS Applied Energy Materials* **4**, 7786 (2021).

21. **S. B. Mishra**, S. Marutheeswaran, S. C. Roy, V. Natarajan, P. K. Rai, and B. R. K. Nanda, "Adsorption and degradation mechanism of 2,4,6-TNT on TiO_2(110) surface," *Surface Science* **713**, 171902 (2021).

22. J. Khatua, T. Arh, **S. B. Mishra**, H. Luetkens, A. Zorko, B. Sana, M. S. R. Rao, B. R. K. Nanda, and P. Khuntia, "Development of short- and long-range magnetic order in double perovskite Ba_2MnTeO_6," *Scientific Reports* **11**, 6959 (2021).

23. **S. B. Mishra**, S. C. Roy, and B. R. K. Nanda, "Electronic structure of graphene/TiO_2 interface: design and functional perspectives," *Applied Surface Science* **542**, 148709 (2021).

24. **S. B. Mishra**, S. K. Yadav, D. G. Kanhere, and B. R. K. Nanda, "Fluorine-intercalated graphene: formation of a two-dimensional spin lattice through pseudoatomization," *Physical Review Materials* **4**, 074411 (2020).

25. **S. B. Mishra** and B. R. K. Nanda, "Facet-dependent catalytic activities of anatase TiO_2 for CO_2 adsorption and conversion," *Applied Surface Science* **531**, 147330 (2020).

26. P. Panigrahi*, **S. B. Mishra***, T. Hussain, B. R. K. Nanda, and R. Ahuja, "Density functional theory studies of Si_2BN nanosheets as anode materials for magnesium ion batteries," *ACS Applied Nano Materials* **3**, 9055 (2020).

27. S. Marutheeswaran*, **S. B. Mishra***, S. C. Roy, and B. R. K. Nanda, "Mechanistic understanding of NO_2 dissociation on a rutile TiO_2(110) surface: an electronic structure study," *Journal of Physical Chemistry C* **124**, 8786 (2020).

28. **S. B. Mishra**, A. Choudhary, S. C. Roy, and B. R. K. Nanda, "Quantum-mechanical process of carbonate complex formation and large-scale anisotropy in the adsorption energy of CO_2 on anatase TiO_2(001) surface," *Physical Review Materials* **2**, 115801 (2018).

29. S. Samanta, **S. B. Mishra**, and B. R. K. Nanda, "Quantum well structure of double perovskite superlattice and formation of spin-polarized two-dimensional electron gas," *Physical Review B* **98**, 115155 (2018).

30. **S. B. Mishra** and B. R. K. Nanda, "Virtual synthesis of crystals using *ab initio* MD: case study on $LiFePO_4$," *AIP Conference Proceedings* **1832**, 090044 (2017).

Manuscripts under review

1. **S. B. Mishra**, F. Belli, E. Zurek, and E. R. Margine, "Anharmonicity and nonadiabaticity in hydride superconductors," arXiv:2608.14949 (2026) [Revision, *Physical Review Letters*].

2. A. Rahman*, **S. B. Mishra***, S. Khan, K. Shah, X. Ma, X. Lin, T. Sobol, L. Zhang, W. Zhao, M. Zhang, and Z. Muhammad, "Symmetry-enforced topology and Van Hove saddle points in hexagonal TiAs," arXiv:2610.12925 (2026).

3. V. Sharma, G. M. Stephen, **S. B. Mishra**, S. Debata, P. Dev, M. Zhukovskyi, T. Orlova, S. P. Bennett, F. M. Able, M. E. Jamer, A. L. Friedman, B. A. Assaf, and D. Heiman, "Topological Hall effect in nanocrystalline FeGe films," [Revision, *Physical Review Materials*].

4. Z. Muhammad*, **S. B. Mishra***, Gayatri, G. Krasucki, W. Ali, K. Olkowska-Pucko, O. Iqbal, Z. U. Rehman, A. U. Rahman, M. R. Molas, L. Xiaoyang, and W. Zhao, "Chiral phonons and giant anisotropic photoresponse in quasi-1D van der Waals semiconductor $ZrSnS_3$," arXiv:2608.22700 (2026) [Revision, *Laser and Photonics Reviews*].

5. S. Tiwari, J. Lafuente-Bartolomé, M. Zacharias, S. Jung, **S. B. Mishra**, T. Y. Kim, K. Yu, Z. Dai, A. Denchfield, J. Park, K. Luo, D. Kim, W. Yang, X. Zhang, B. Cucco, V.-A. Ha, W. G. Vandenberghe, S. Poncé, E. Kioupakis, E. R. Margine, and F. Giustino, "Programmable electron-phonon and many-body workflows with EPWpy," [Revision, *npj Computational Materials*].

6. **S. B. Mishra**, S. ElMeligy, and P. Dev, "Superconductivity in noncentrosymmetric NbReSi: beyond harmonic and adiabatic limits," arXiv:2609.30564 (2026) [Submitted to *Physical Review B*].

7. Akshat Singla, **S. B. Mishra**, Nimai Pathak, Manjud, and Anup Thakur, "Tuning Band Gap and NIR Photoluminescence via Sr/Ba Substitution in K(Sr, Ba)$ScSi_2O_7$," [Submitted to *Journal of Materials Chemistry A*].

Manuscripts in preparation

1. S. Jois, **S. B. Mishra**, J. Dong, J. L. Spellberg, T. Xie, T. Alem, K. Burns, G. Gu, V. Kulichenko, L. Balicas, A. J. Sternbach, C. Gong, G. M. Stephen, P. Dev, A. T. Hanbicki, and A. L. Friedman, "Insulator-like behavior in $Bi_2Sr_2CaCu_2O_{8+x}$ driven by Fe_5GeTe_2 at room temperature".

2. N. W. G. Smith, **S. B. Mishra**, S. Jois, P. Dev, G. M. Stephen, S. Krylyuk, A. Davydov, L. Balicas, A. T. Hanbicki, and A. L. Friedman, "Magnetic-field-assisted resistive switching in α-In_2Se_3/Fe_5GeTe_2 heterostructures".

3. **S. B. Mishra**, S. Debata, and P. Dev, "Raising the magnetic ordering scale of B20 FeGe: surfaces, lattice distortion, and local chemistry from first principles."

4. **S. B. Mishra**, I. Naumov, and P. Dev, "Strain induced superconductivity dome in 2M-WS_2."

5. C. Dawra, **S. B. Mishra**, Z. Liu, and E. R. Margine, "Comparative study of phonon-limited carrier transport in the nodal-line semimetal ZrSiX (X = S, Se, Te) family."

RESEARCH GRANTS

- **Principal Investigator**, ACCESS Allocation PHY260334 (2026–2027): 150,000 NSF service-unit hours for first-principles simulations of quantum materials.
- **Principal Investigator**, ACCESS Allocation PHY250052 (2025–2027): 150,000 NSF service-unit hours for van Hove singularities in H_3S.

HONORS AND AWARDS

- APS DCOMP Travel Award (2026), American Physical Society, Global Physics Summit, Denver, Colorado.
- Young Scientist Award (2024), Odisha Physical Society, India.
- Best Ph.D. Thesis Award in Physics (2021), Prof. A. L. Lashkar Prize, IIT Madras, India.
- DST Young Scientist Travel Award (2019), Govt. of India, for ICAMM, Rennes, France.
- Oral Presentation Award, EESTER International Conference, SRM and IIT Madras (2020), India.
- Best Poster Award, Conference on Advanced Functional Materials (ICAFM), Anna University (2017), India.
- First Prize, I2CAM School on Clean and Renewable Energy, JNCASR, Bangalore (2017), India.
- Meritorious Master Scholarship, Institute of Mathematics & Applications, DST India (2011–2013).

TEACHING EXPERIENCE

Guest Lecturer, Statistical Mechanics I (PHYS 631), graduate course, Department of Physics, Binghamton University Fall 2025

Teaching Assistant, Department of Physics, IIT Madras 2015–2019
Undergraduate and graduate lecture and laboratory courses (Physics I/II, Engineering and M.Sc. Physics Labs).

Workshop and Short-Course Instructor

- Instructor, APS Short Course: *Electron-Phonon Physics from First Principles: Theory and Computational Tools*, APS Global Physics Summit, Denver, CO (2026).
- Instructor, EPW–Quantum ESPRESSO Summer School on Electron-Phonon Physics, UT Austin, TX (2024).
- Instructor, *EESTER–2018 Workshop* on Quantum ESPRESSO and first-principles methods, jointly organized by SRM IST and IIT Madras, India (2018).

Teaching Interests

- **Graduate:** Quantum Mechanics, Statistical Mechanics, Electrodynamics, Condensed Matter, Many-Body Theory.
- **Undergraduate:** Introductory Physics and lab, Solid State Physics, Mathematical Methods, Computational Physics.
- **Electives:** Magnetism and Quantum Materials, First-Principles Electronic Structure, Electron–Phonon Physics.

MENTORING

- **Ph.D. students co-mentored (current):** Pawan Joshi (SUNY Binghamton), Sohair ElMeligy (Virginia Tech), and Benjamin Edun (University of Maryland), on superconductivity and silicon defects and interfaces.
- **Ph.D. students co-mentored (previous):** Abhijitha V. G. (IIT Madras), on Al-ion battery materials, resulting in four co-authored publications.
- **Integrated M.Sc. students projects (IIT Madras):** Harshvardhan Murali and Sajid Ali, on magnetism in defected and halogen-intercalated graphene.

SOFTWARE AND WORKFLOW DEVELOPMENT

- Contributed to EPW and EPWpy, with workflows for electron–phonon coupling and superconductivity.
- Developed magnetism workflows linking DFT to exchange, anisotropy, and Monte Carlo ordering temperatures.
- Implemented Weyl-node detection, adaptive Berry-curvature sampling, Berry-curvature corrections to Boltzmann transport, and two-level MPI parallelization of the EPW superconductivity module.
- Derived and implemented the inverse Faraday effect for metals in WannierBerri.
- Contribute to superconductivity EPW documentation and tutorials.

PROFESSIONAL SERVICE

Leadership in Professional Societies

- Secretary/Treasurer, APS Topical Group on Energy Research and Applications (GERA) (2025–2027).
- Early Career Representative, APS Division of Computational Physics (DCOMP) (2024–2025).
- APS Annual Leadership Meeting and Congressional Visit: Representative for DCOMP, Washington, DC (2025).

Conference and Session Organization

- Session Chair, APS Global Physics Summit 2026, Session S46: "Exciton–Phonon Interactions", Denver, CO.
- Session Chair, APS March Meeting 2024, Session K58: "Electrons, Phonons, and Electron-Phonon Scattering".
- Participant and presenter, NSF Cyberinfrastructure for Sustained Scientific Innovation (CSSI) PI Meeting, Denver, CO (July 2025).

Peer Review

- Reviewed manuscripts for journals including *Phys. Rev. Lett.*, *Phys. Rev. Applied*, *Phys. Rev. B*, *Appl. Phys. Lett.*, *Langmuir*, *ACS Appl. Mater. Interfaces*, *Sci. Rep.*, *Comput. Phys. Commun.*, and *J. Energy Storage*.

Professional Memberships

- Member, Sigma Xi: The Scientific Research Honor Society (University of Maryland Chapter, 2026–present).
- Member, American Physical Society (DCOMP, DCMP, GERA) (2021–present).

CONFERENCE PRESENTATIONS

Invited Talks

- "Insights into the surface chemistry of oxide semiconductors," *ICAM-SEEi Workshop*, IITRAM, Ahmedabad, India (2025).
- "Understanding phonon-mediated superconductivity: from fundamentals to frontiers," *Faculty Development Program on Frontiers in Material Science*, SRM IST, Chennai, India (2025).
- "Phonon-mediated superconductivity in H_3S and NaC_4," *Young Scientist Award Talk, Orissa Physical Society*, KIIT, Bhubaneswar, India (2025).
- "An introduction to electronic properties of materials and magnetism in graphene," *Physics Seminar, Fakir Mohan College*, Balasore, India (2019).

Selected Contributed Talks (from 30 total)

- "Anharmonic and nonadiabatic electron-phonon effects in hydride superconductors," *APS Global Physics Summit*, Denver (2026, accepted).
- "Automating many-body workflows with EPWpy in the MATCSSI framework," *NSF CSSI-PI Meeting*, Denver (2025).
- "Stability–superconductivity map for compressed Na-intercalated graphite," *APS March Meeting*, Anaheim (2025).
- "Phonon-mediated superconductivity beyond the Migdal approximation," *APS March Meeting*, Minneapolis (2024).
- "Theory of inverse Faraday effect in $3d$, $4d$, and $5d$ metals," *APS March Meeting*, Las Vegas (2023).

COMPUTATIONAL EXPERTISE

- **Electronic structure & magnetism:** DFT, DFPT, DFT+U, spin–orbit coupling, noncollinear magnetism, magnetic symmetry, exchange and anisotropy, spin models, Monte Carlo.
- **Superconductivity & lattice dynamics:** Electron–phonon coupling, Eliashberg theory, vertex corrections, anharmonic phonons, AIMD, machine-learned force fields.
- **Topology & transport:** Wannier interpolation, Berry phases, topological invariants, Boltzmann transport, Hall responses.
- **Codes & HPC:** Quantum ESPRESSO, EPW/EPWpy, VASP, ABINIT, Wannier90, WannierBerri, WannierTools, IRVSP; Fortran, Python, MPI/OpenMP, SLURM.